

# PASSIVE KEYLESS ENTRY

## High-Level Design Document

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Classification: Synthetic Demonstration Data

## 1. System Overview

The Passive Keyless Entry coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

## 2. Software Architecture

The primary software components are AccessManager, KeyFobControl, DoorLockControl, ImmobilizerControl, and ProximitySensorManager.

## 3. Software Components

| Component              | Role        | Primary Responsibility                 |
|------------------------|-------------|----------------------------------------|
| AccessManager          | Coordinator | Authorises entry and engine start.     |
| KeyFobControl          | Provider    | Receives and decodes key fob messages. |
| DoorLockControl        | Consumer    | Locks and unlocks the doors.           |
| ImmobilizerControl     | Consumer    | Releases engine start authorisation.   |
| ProximitySensorManager | Provider    | Estimates the key distance.            |

## 4. Interfaces

| Interface     | Provider               | Consumer           | Signals            |
|---------------|------------------------|--------------------|--------------------|
| IKeyMessage   | KeyFobControl          | AccessManager      | KeyChallenge       |
| IKeyDistance  | ProximitySensorManager | AccessManager      | KeyDistance        |
| ILockRequest  | AccessManager          | DoorLockControl    | LockState          |
| IStartRelease | AccessManager          | ImmobilizerControl | StartAuthorisation |

## 5. Ports

| Component              | Port             | Direction | Interface          |
|------------------------|------------------|-----------|--------------------|
| KeyFobControl          | KeyMessage_Out   | P-PORT    | IKeyMessage        |
| AccessManager          | KeyMessage_In    | P-PORT    | IKeyMessage        |
| ProximitySensorManager | KeyDistance_Out  | P-PORT    | IKeyDistance       |
| AccessManager          | KeyDistance_In   | R-PORT    | IKeyDistance       |
| AccessManager          | LockRequest_Out  | P-PORT    | ILockRequestLegacy |
| DoorLockControl        | LockRequest_In   | R-PORT    | ILockRequest       |
| AccessManager          | StartRelease_Out | P-PORT    | IStartRelease      |
| ImmobilizerControl     | StartRelease_In  | R-PORT    | IStartRelease      |

## 6. Signals

| Signal             | Data Type | Source                 | Destination     |
|--------------------|-----------|------------------------|-----------------|
| KeyChallenge       | uint32    | KeyFobControl          | AccessManager   |
| KeyDistance        | uint8     | ProximitySensorManager | AccessManager   |
| LockState          | boolean   | AccessManager          | DoorLockControl |
| StartAuthorisation | boolean   | AccessManager          | KeyFobControl   |

## 7. Dependencies

| Source                 | Relationship       | Target        | Reason                         |
|------------------------|--------------------|---------------|--------------------------------|
| KeyFobControl          | PROVIDES_INTERFACE | IKeyMessage   | Publishes key challenge.       |
| AccessManager          | REQUIRES_INTERFACE | IKeyMessage   | Consumes key challenge.        |
| ProximitySensorManager | PROVIDES_INTERFACE | IKeyDistance  | Publishes key distance.        |
| AccessManager          | PROVIDES_INTERFACE | ILockRequest  | Publishes lock state.          |
| DoorLockControl        | REQUIRES_INTERFACE | ILockRequest  | Consumes lock state.           |
| AccessManager          | PROVIDES_INTERFACE | IStartRelease | Publishes start authorisation. |
| ImmobilizerControl     | REQUIRES_INTERFACE | IStartRelease | Consumes start authorisation.  |

## 8. Functional Flows

### Flow F-001: Key Challenge Distribution

KeyFobControl publishes KeyChallenge through IKeyMessage. AccessManager receives the signal through KeyMessage\_In and updates its internal state model.

### Flow F-002: Key Distance Distribution

ProximitySensorManager publishes KeyDistance through IKeyDistance. AccessManager receives the signal through KeyDistance\_In and updates its internal state model.

### Flow F-003: Lock State Distribution

AccessManager publishes LockState through ILockRequest. DoorLockControl receives the signal through LockRequest\_In and updates its internal state model.

### Flow F-004: Start Authorisation Distribution

AccessManager publishes StartAuthorisation through IStartRelease. ImmobilizerControl receives the signal through StartRelease\_In and updates its internal state model.

## 9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

## 10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.